



OPEN

Optimized decomposition and deep learning with bias correction for reliable runoff point-interval prediction

Hong Ma^{1,2}, Muhammad Fadhil Marsani²✉, Mohd. Asyraf Mansor³,
Mohd Shareduwan Mohd Kasihmuddin² & Caiyan Long²

Accurate runoff prediction is critical for flood risk management and water resources regulation. This study proposes a probabilistic runoff forecasting framework that integrates optimized signal decomposition, deep learning, bias correction, and uncertainty quantification. Variational Mode Decomposition optimized by the Whale Optimization Algorithm (WVMD) is first applied to decompose non-stationary runoff series into stable intrinsic mode functions, which are then modeled using a hybrid Temporal Convolutional Network and Bidirectional Gated Recurrent Unit (TCN-BiGRU) with optimized hyperparameters. A Bias Correction (BC) strategy is further incorporated to improve point prediction accuracy. Based on the runoff point prediction residuals generated by the WVMD-TCN-BiGRU-BC model, Kernel Density Estimation (KDE) is applied to characterize the error distribution and construct probabilistic prediction intervals, with the Normal and Gumbel distributions adopted as parametric benchmarks. Experimental results at the Yangtze River Basin demonstrate that the proposed framework significantly improves both deterministic and probabilistic forecasting performance. The BC strategy reduces RMSE by 68.0% at Yichang and 73.2% at Jianli station compared with the WVMD-TCN-BiGRU model, while the KDE based interval prediction yields an approximately 5% improvement in the F score at the 90% confidence level, confirming the reliability of the proposed runoff probabilistic forecasting framework.

Keywords Temporal convolutional network, Bidirectional gated recurrent unit, Whale optimization algorithm, Bias correction, Kernel density estimation

Flood forecasting plays a crucial role in the effective allocation of water resources and the optimal management of river basins. The Yangtze River Basin, influenced by a subtropical monsoon climate, frequently experiences heavy rainfall and flooding, particularly during the summer months from June to August. Such events pose severe threats to public safety and property due to their sudden onset and extreme intensity^{1–3}. As hydrological monitoring stations along the Yangtze River play a critical role in flood forecasting, their forecast data are essential for early warning, disaster prevention, and risk management under extreme weather conditions. However, unpredictable climate change and rapid infrastructure development have substantially altered hydrological patterns within the basin, presenting significant challenges for accurate flood and runoff forecasting in high-risk regions⁴.

With advances in artificial intelligence⁵, deep learning (DL) techniques have been increasingly applied in hydrological forecasting⁶. Nevertheless, the performance of DL models is often degraded by the presence of non-stationary characteristics and multi-scale temporal features in runoff data. Consequently, data preprocessing has become a critical step in improving forecasting accuracy^{7,8}. Previous studies have shown that combining ensemble empirical mode decomposition (EEMD) with Long Short-Term Memory (LSTM) networks can significantly enhance predictive performance^{9–11}. Variational Mode Decomposition (VMD), a non-recursive signal decomposition technique, further alleviates mode mixing by introducing penalty factors and bandwidth constraints¹². Integrating VMD with deep learning models such as CNN, LSTM, and GRU has demonstrated notable improvements in forecasting accuracy^{13–16}. However, CNN-based models often struggle to preserve

¹School of Financial Mathematics and Statistics, Guangdong University of Finance, Guangzhou 510521, China.

²School of Mathematical Sciences, Universiti Sains Malaysia, Penang 11800, Malaysia. ³School of Distance Education, Universiti Sains Malaysia, Penang 11800 USM, Malaysia. ✉email: fadhilmarsani@usm.my

temporal order in tasks with strong sequential dependence. To overcome this limitation, Temporal Convolutional Networks (TCN) were proposed by Bai et al.¹⁷, employing causal and dilated convolutions to effectively capture long-term temporal dependencies with improved computational efficiency. In addition, Bidirectional Gated Recurrent Units (BiGRU) address the unidirectional dependencies in time series and help overcome the limitations of LSTM and GRU models. When combined with TCN, BiGRU-based models have shown enhanced forecasting performance across various application domains^{18–23}. Despite these advances, model performance remains highly sensitive to hyperparameter settings²⁴. Additionally, both the number of modes and penalty parameters in VMD affect decomposition quality and forecasting accuracy. Therefore, effective hyperparameter optimization is essential to fully exploit the synergy among model components.

The use of intelligent algorithms to optimize model parameters is seen to improve prediction accuracy²⁵. There is evidence that the Whale Optimization Algorithm (WOA)²⁶ plays a crucial role in effectively regulating model parameters and is considered a good choice due to its simple structure and comprehensive global search capabilities. Zhao et al.²⁷ optimized a BiLSTM model using WOA for short-term electricity load forecasting, achieving improved accuracy and reduced parameter tuning. Chen et al.²⁸ applied WOA to optimize a BiLSTM-AM model for wind power forecasting, showing improved performance over non-optimized counterparts. In flood prediction, Ji et al.²⁹ optimized a GRU model using WOA, achieving improved performance in time series forecasting. In this study, WOA-optimized VMD, named WVMD, is utilized to determine the parameters of VMD. WOA is also employed to tune the hyperparameters of the TCN-BiGRU model. However, several important questions remain regarding datasets that often contain data noise and model limitations. Data noise, which refers to random variations or errors in the dataset, can lead to inaccurate model training and predictions, as it introduces unnecessary complexity that obscures the true underlying patterns^{30,31}. Such noise can arise from measurement errors, environmental variations, or inconsistencies in data collection processes. Model limitations, on the other hand, occur when the model is unable to fully capture the complexity of the data, either due to oversimplified assumptions, insufficient training data, or inadequate model architecture. These factors can result in poor generalization, over-fitting, or under-fitting, all of which negatively impact the model's predictive performance. Hence, this study provides an account of DL-based bias correction (BC) by introducing the BC strategy into the WVMD-TCN-BiGRU, resulting in the WVMD-TCN-BiGRU-BC model for runoff forecasting.

Uncertainty modeling is essential for enhancing the reliability of hydrological forecasts. Interval forecasting provides a range of possible outcomes that explicitly accounts for uncertainty arising from environmental variability and data noise, thereby supporting more informed decision making. Probability-based interval forecasting sets confidence intervals to determine the upper and lower bounds of the forecast range, offering more valuable references for decision-makers^{32,33}. Although numerous probabilistic approaches have been developed for streamflow forecasting^{9,34}, traditional parametric probability distribution forecasting requires assumptions about error distributions that may not hold in practice³⁵. Moreover, many existing studies focus primarily on point forecasting, with limited investigation into how probabilistic interval prediction influences overall forecasting performance. To address this gap, this study employs nonparametric Kernel Density Estimation (KDE) to model forecast error distributions and generate flexible probabilistic prediction intervals. Since KDE performance is highly sensitive to bandwidth selection, its effectiveness is evaluated by comparison with traditional parametric distributions, including the Normal and Gumbel distributions, based on point runoff prediction errors.

This study advances beyond a simple hybridization of existing techniques by introducing a dual-level, WOA-driven optimization framework that explicitly and tightly integrates signal decomposition, point prediction, and probabilistic interval estimation into a unified forecasting pipeline.

The contributions of this study are as follows:

- i. Introducing the WVMD method to decompose runoff data into high quality IMFs, significantly enhancing data stationarity and reducing noise. This approach improves forecast quality by providing more refined inputs for modeling, leading to more accurate predictions.
- ii. Developing a novel hybrid deep learning model, WVMD-TCN-BiGRU-BC, which integrates the strengths of TCN and BiGRU. The TCN effectively captures long-term temporal dependencies through causal dilated convolutions, while the BiGRU leverages bidirectional processing to account for both past and future context in time series data. The hyperparameters of the model are optimized using the WOA, resulting in highly accurate runoff forecasts.
- iii. Incorporating a BC mechanism into the framework, which utilizes forecast error series to refine initial point predictions. This method minimizes systematic bias and enhances the overall accuracy of runoff forecasting, thereby establishing a more robust and reliable prediction system.
- iv. Applying the ROT-KDE method to model the probability density of forecast error sequences using non-parametric estimation. By comparing with parametric methods (Normal and Gumbel distributions), this approach generates more reliable prediction intervals, enabling a quantitative assessment of uncertainty and improved decision support in water resource management.

Runoff point prediction methods Optimized variational mode decomposition

Whale optimization algorithm

The Whale Optimization Algorithm (WOA), proposed by Mirjalili³⁶, is a population-based metaheuristic inspired by the bubble-net hunting behavior of humpback whales. It incorporates three main strategies: encircling prey, spiral predation, and a random search mechanism.

Step 1. Encircling Prey.

Whales update their positions relative to the best solution found so far, simulating a shrinking encircling mechanism:

$$\begin{cases} X(t+1) = X_{best}(t) - A \cdot D_1 \\ D_1 = |C \cdot X_{best}(t) - X(t)| \\ A = 2a \cdot r_1 - a, C = 2r_2 \end{cases} \quad (1)$$

where $X(t)$ denotes the current position of the whale, and $X_{best}(t)$ represents the current optimal position of the prey. A and C are coefficient vector, where a decreases linearly from 2 to 0 during the optimization process, and r_1, r_2 are random numbers within the range $[0, 1]$. When $|A| < 1$, the whale moves toward the optimal solution, simulating a shrinking encircling behavior.

Step 2. Global Search.

To enhance exploration and avoid premature convergence, when $|A| \geq 1$, whales randomly select a position $X_{rand}(t)$ rand from the current population:

$$\begin{cases} X(t+1) = X_{rand}(t) - A \cdot D_2 \\ D_2 = |C \cdot X_{rand}(t) - X(t)| \end{cases} \quad (2)$$

Step 3. Spiral Predation.

Whales approach the prey along a logarithmic spiral path, which is modeled as:

$$\begin{cases} X(t+1) = D_3 \cdot e^{bl} \cdot \cos(2\pi l) + X_{best}(t) \\ D_3 = |X_{best}(t) - X(t)| \end{cases} \quad (3)$$

where D_3 represents the distance between the whale and its prey, b controls the spiral contraction rate, and l is a random number uniformly distributed in the range $[-1, 1]$.

Step 4. Probability-based Strategy.

With probability, whales choose between the encircling mechanism and the spiral updating strategy:

$$X(t+1) = \begin{cases} X_{best}(t) - A \cdot D_1, & \text{if } P < 0.5 \text{ and } |A| < 1 \\ X_{rand}(t) - A \cdot D_2, & \text{if } P < 0.5 \text{ and } |A| \geq 1 \\ D_3 \cdot e^{bl} \cdot \cos(2\pi l) + X_{best}(t), & \text{if } P \geq 0.5 \end{cases} \quad (4)$$

To comprehensively evaluate the performance of WOA in solving complex optimization problems, six representative benchmark functions (F1-F6) from the CEC2021 test suite were selected for comparison. These functions include unimodal, multimodal, hybrid, and composition problems, providing a balanced assessment of both global exploration and local exploitation capabilities. The experimental settings were as follows: population size of 30, maximum number of iterations of 500, and problem dimension of 10. The comparative algorithms included Particle Swarm Optimization (PSO), Goose Optimization Algorithm (GO), Genetic Algorithm (GA), Bitterling Fish Optimization (BFO), Escape Algorithm (ESC), and Differential Creative Search (DCS).

As shown in Fig. 1, the convergence curves illustrate the variation of the best fitness values across iterations for each function. Evidently, WOA outperforms other algorithms in all six test functions. For F1 and F2, WOA quickly approaches the theoretical optimum in early iterations, with a much faster convergence speed than PSO, GA, and BFO. In F3, WOA has a smooth and stable descent in nonlinear landscapes, indicating strong local exploitation ability. For F4 and F5, WOA achieves consistent optimization without premature convergence or oscillatory behavior, highlighting its robustness in complex environments. In F6, WOA keeps improving performance under periodic and irregular search conditions, showing excellent global exploration capability.

Overall, WOA has faster convergence, higher solution accuracy, and better stability than competing algorithms.

Variational mode decomposition

Variational Mode Decomposition (VMD) is an advanced time-frequency analysis technique designed for effectively handling nonlinear and non-stationary signals. VMD decomposes a given signal $x(t)$ into K intrinsic mode functions (IMFs) by solving a constrained variational problem formulated as follows:

$$\begin{cases} h(t) = (\delta(t) + \frac{j}{\pi t}) * u_k(t) \\ \min_{\{u_k\}, \{\omega_k\}} \left(\sum_k \|\partial_t h(t) e^{-j\omega_k t}\|_2^2 \right) \\ s.t. \sum_k u_k = x(t) \end{cases} \quad (5)$$

where $h(t)$ denotes the Hilbert transform, $\delta(t)$ represents the Dirac delta function, $*$ represents the convolution operator, u_k is denoted as the k -th IMF, and ω_k is its corresponding center frequency, ∂_t is the first-order partial derivative with respect to time t . The term $e^{-j\omega_k t}$ is the spectral shift of u_k to the exponential term of ω_k .

By introducing a quadratic penalty factor α and a Lagrange multiplier λ , the constrained variational problem in Eq. (5) is transformed into an unconstrained optimization problem:

$$L(\{u_k\}, \{\omega_k\}, \lambda) = \alpha \sum_k \|\partial_t h(t) e^{-j\omega_k t}\|_2^2 + \|x(t) - \sum_k u_k\|_2^2 + \langle \lambda(t), x(t) - \sum_k u_k \rangle \quad (6)$$

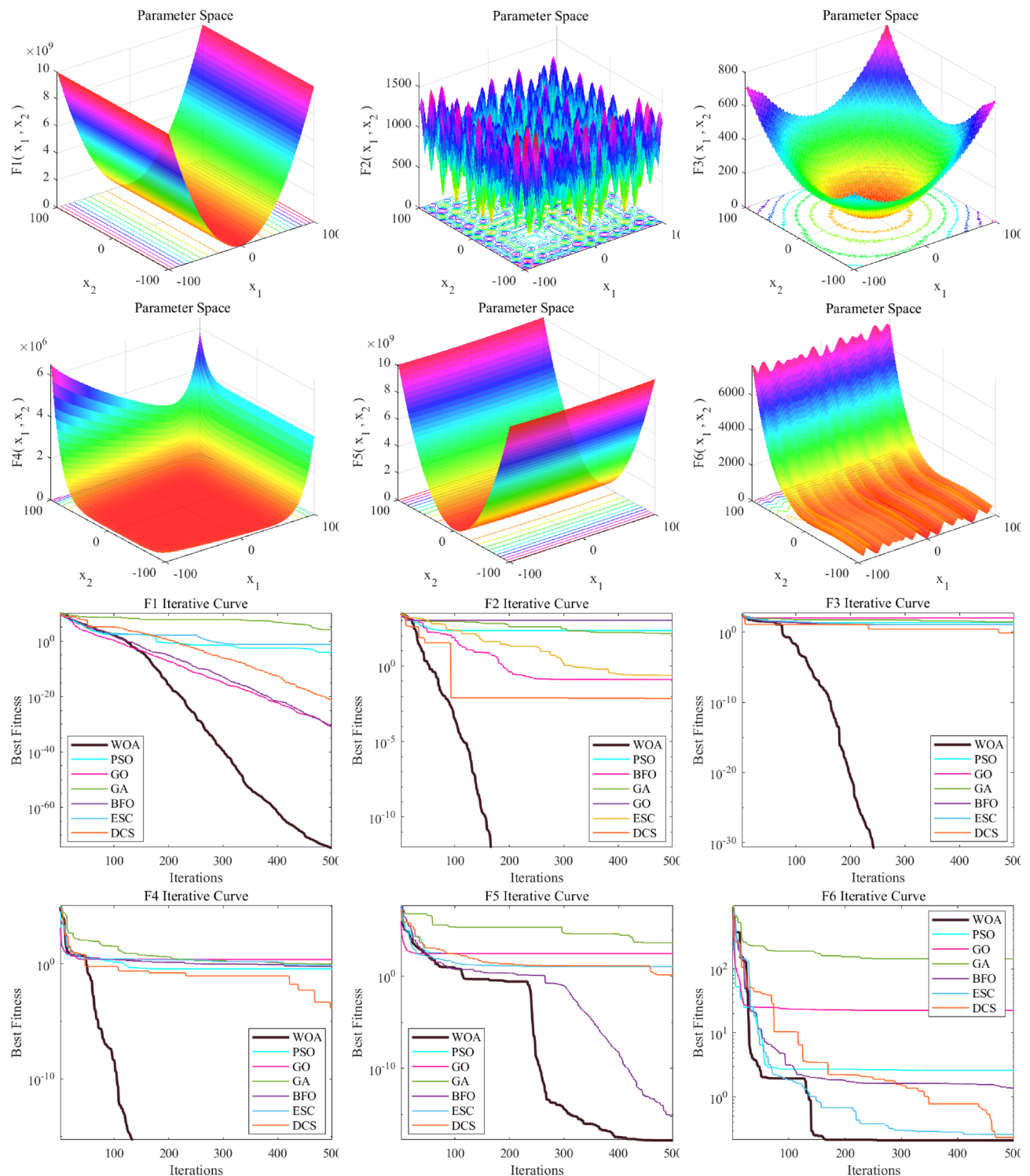


Fig. 1. Convergence curves of WOA and competing algorithms on the CEC2021 benchmark functions (F1-F6).

The time-frequency transformation in Eq. (6) is performed using the Alternating Direction Method of Multipliers (ADMM), which iteratively identifies the saddle point of the augmented Lagrangian formulation, as given by:

$$\begin{cases} \hat{u}_k^{n+1}(\omega) = \frac{\hat{x}(\omega) - \sum_{i \neq k} \hat{u}_i(\omega) + \frac{\hat{\lambda}(\omega)}{2}}{1 + 2\alpha(\omega - \omega_k)^2} \\ \hat{\omega}_k^{n+1} = \frac{\int_0^\infty \omega |\hat{u}_k(\omega)|^2 d\omega}{\int_0^\infty |\hat{u}_k(\omega)|^2 d\omega} \\ \hat{\lambda}^{n+1}(\omega) = \hat{\lambda}^n(\omega) + \tau (\hat{x}(\omega) - \sum_k \hat{u}_k^{n+1}(\omega)) \end{cases} \quad (7)$$

where $\hat{x}(\omega)$, $\hat{u}_k(\omega)$ and $\hat{\lambda}(\omega)$ denote the Fourier transforms of $x(t)$, $u_k(t)$ and $\lambda(t)$, respectively; ω is the angular frequency; n represents the iteration index; and τ is the fidelity parameter.

WOA-optimized VMD (WVMD)

The performance of VMD is sensitive to two parameters: the penalty factor α , which controls the bandwidth smoothness of each mode, and the number of modes K , which determines the decomposition resolution. In this study, these parameters are automatically optimized using the WOA, as illustrated in Fig. 2.

The parameter optimization task is formulated as a continuous minimization problem. Each candidate solution (K, α) is evaluated by applying VMD to the runoff series and computing a permutation entropy (PE) based objective function. Specifically, PE is employed to quantify the regularity and complexity of the decomposed IMFs. For each candidate parameter pair, VMD is first applied to the runoff data, and the PE of each resulting IMF is calculated. The minimum PE value among all IMFs is selected as the fitness value. This strategy guides the optimization toward parameter combinations that produce at least one highly structured, low-complexity mode with strong predictive potential.

The search ranges of the parameters are set to $\alpha \in [100, 2500]$ and $K \in [3, 10]$. The WOA is initialized with a population size of 25 and executed for 20 iterations. During each iteration, all candidate solutions are restricted within the predefined bounds. The candidate yielding the lowest PE value is regarded as the current leader. The algorithm updates all candidate positions using standard WOA mechanisms to balance global exploration and local exploitation. Subsequently, the positions of all candidates are updated using WOA to balance global exploration and local exploitation.

Throughout the optimization process, two key records are maintained:

- The convergence curve, which monitors the evolution of the best PE value over iterations.

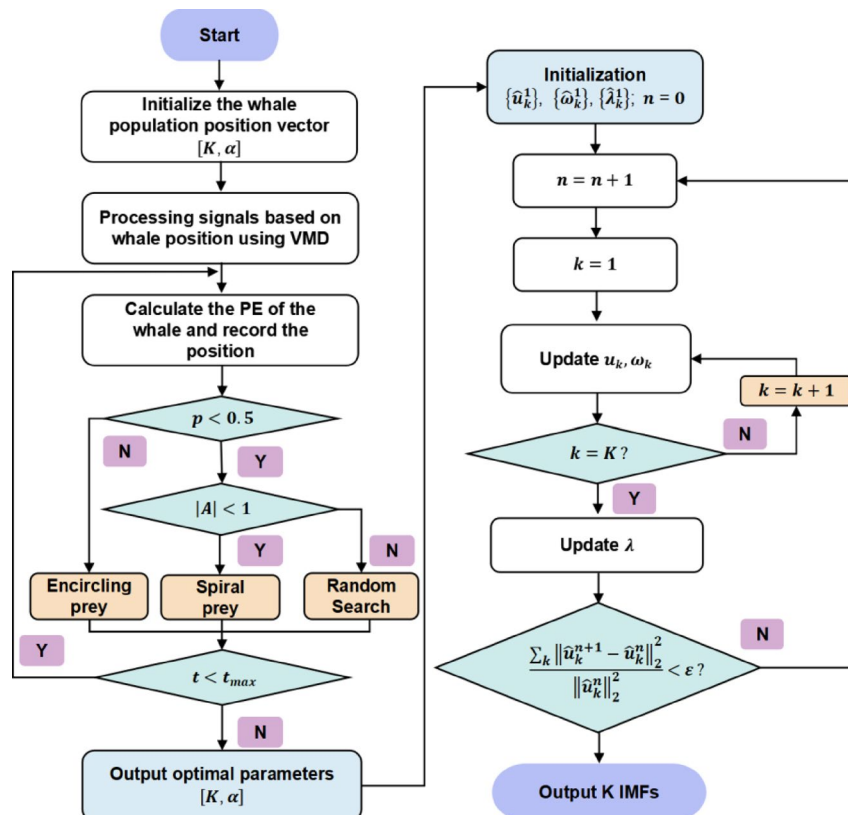


Fig. 2. Flowchart of WOA-optimized VMD (WVMD).

- ii. The best-parameter history, which tracks the optimal α and K values of the leader solution throughout the optimization process.

Upon convergence, the final leader position (K, α) is selected as the optimal VMD configuration. Subsequently, the optimized parameters are employed to decompose the runoff series. This process generates cleaner IMFs with less mode mixing and then feeds them into the downstream prediction model.

Optimized TCN-BiGRU networks with bias correction

Temporal convolutional networks

The Temporal Convolutional Network (TCN) integrates the strengths of convolutional neural networks (CNNs) and recurrent neural networks (RNNs), while effectively avoiding information leakage commonly encountered in conventional CNN-based sequence modeling³⁷. For one-dimensional time-series inputs, TCN employs dilated causal convolutions, which expand the receptive field by adjusting the dilation factor without increasing the number of parameters. The dilated convolution operation is defined as:

$$F(s) = \sum_{j=0}^{k-1} f(j) \cdot x_{s-d \cdot j} \quad (8)$$

where $f(\cdot)$ represents the filter function, and $s - d \cdot j$ indicates the direction of past information. TCN adopts residual connections to facilitate deep network training. Each residual block consists of two stacked causal dilated convolutional layers, and a 1×1 convolutional layer is added after the first convolution block to match dimensions. These residual connections enhance the receptive field of networks, enable deeper training, and alleviate gradient vanishing and explosion issues. In the proposed model, layer normalization is applied after each causal dilated convolution to stabilize training, and the Rectified Linear Unit (ReLU) activation function is used to introduce non-linearity, enhancing the ability of the model to capture complex patterns. Additionally, dropout is employed to reduce overfitting. The overall TCN structure is illustrated in Fig. 3(a).

Bidirectional gated recurrent units

The Bidirectional Gated Recurrent Units (BiGRU) consist of two GRU networks that process the input sequence in forward and backward temporal directions³⁸, respectively. The internal structure of the BiGRU is shown in Fig. 3(b). The GRU is a simplified version of the Long Short-Term Memory (LSTM) network featuring a more efficient gating mechanism and fewer training parameters. This reduction in parameters makes the computation process more efficient, thereby improving the speed of sample training. The computation process of a GRU unit is described as:

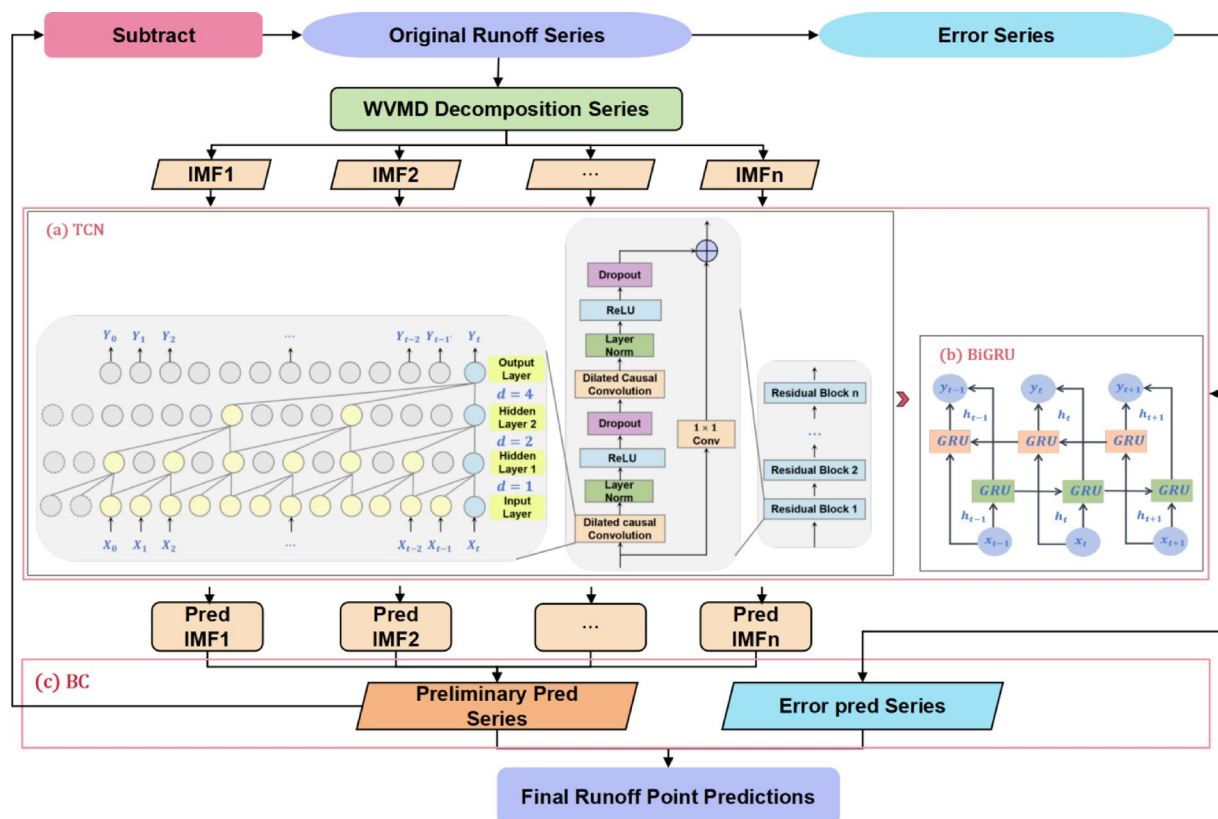


Fig. 3. Framework of the WVMD-TCN-BiGRU-BC runoff point prediction model.

$$\begin{cases} r_t = \sigma(W_{rx}x_t + W_{rh}h_{t-1} + b_r) \\ z_t = \sigma(W_{zx}x_t + W_{zh}h_{t-1} + b_z) \\ \tilde{h}_t = \tanh\left(W_{\tilde{h}x}x_t + W_{\tilde{h}h}(r_t \otimes h_{t-1}) + b_{\tilde{h}}\right) \\ h_t = (1 - z_t) \otimes h_{t-1} + z_t \otimes \tilde{h}_t \end{cases} \quad (9)$$

where x_t and h_t denote the input vector and hidden state at time t ; r_t and z_t correspond to the reset gate and update gate, σ represents the sigmoid function, h_t is the parameter to be updated, $\tanh$ is the activation function. The symbols $\otimes$ and $\oplus$ denote element-wise multiplication and addition, respectively, while W and b represent the weight and bias.

At time t , the BiGRU output is jointly determined by three components: the bias term, the forward GRU_f output $\vec{h}_t$, and the backward GRU_b output $\overleftarrow{h}_t$. The corresponding formulation is expressed as follows:

$$\begin{cases} \vec{h}_t = GRU_f(x_t, \vec{h}_{t-1}) \\ \overleftarrow{h}_t = GRU_b(x_t, \overleftarrow{h}_{t-1}) \\ h_t = V_t \vec{h}_t + W_t \overleftarrow{h}_t + b_t \end{cases} \quad (10)$$

where, x_t represents the input, $\vec{h}_t$ and $\overleftarrow{h}_t$ respectively represent the forward and backward outputs, V_t , W_t and b_t represent the weights and biases.

WOA optimized TCN-BiGRU

The proposed TCN-BiGRU model is designed to jointly exploit multi-scale temporal feature extraction and bidirectional dependency modeling for runoff prediction. In this hybrid architecture, a TCN and a BiGRU are integrated in a sequential parallel manner to fully capture both local and long-range temporal patterns.

Specifically, the input runoff sequence is first processed by a multi-layer TCN composed of dilated causal convolution blocks with residual connections. By progressively increasing the dilation factors, the TCN effectively captures temporal features at multiple time scales while strictly preserving temporal causality. The extracted high-level temporal representations are then fed into the BiGRU network, which models bidirectional temporal dependencies by jointly learning forward and backward sequence information. This structure enables the hybrid model to capture complex nonlinear dynamics in runoff series with enhanced representation capacity.

Following the establishment of the TCN-BiGRU prediction mechanism, the WOA is further employed at the prediction stage to automatically search for the optimal hyperparameter configuration of the hybrid model. Inspired by the bubble-net hunting strategy of humpback whales, WOA updates the positions of search agents through encircling prey, spiral updating, and random search, with the root mean squared error (RMSE) between predicted and observed runoff serving as the objective function.

Four key hyperparameters are optimized, including the number of convolutional filters, the convolutional filter size, the dropout rate, and the initial learning rate.³ The corresponding search ranges are set to values of 8 to 64, 3 to 8, 0.001 to 0.3, and 0.001 to 0.01, respectively. The population size and the maximum number of iterations are configured as 30 and 50. During each iteration, candidate hyperparameter configurations are evaluated, and WOA dynamically updates the search positions based on the best-performing solution, gradually converging toward the global optimum.

The optimal hyperparameter set obtained via WOA is subsequently adopted to train the TCN-BiGRU model. This adaptive optimization strategy effectively balances feature extraction capability, model complexity, and predictive accuracy, thereby enhancing the robustness and overall performance of the proposed runoff forecasting framework.

Bias correction

The bias correction (BC) strategy proposed in this study consists of three main steps.

Step 1. To reduce the complexity of the original runoff sequence, the WVMD method is first employed to decompose the runoff sequence into several relatively stable IMFs. The TCN-BiGRU model is applied to predict each IMF obtained from the WVMD decomposition. The predicted IMFs are subsequently summed to generate the initial runoff prediction sequence y_{I_pre} .

Step 2. The prediction error sequence y_{Error} is calculated as the difference between the actual runoff series y_{Act} and the initial predicted values y_{I_pre} obtained in Step 1:

$$y_{Error} = y_{Act} - y_{I_pre} \quad (11)$$

Step 3. The TCN-BiGRU model is further employed to model the error sequence y_{Error} . The predicted error values y_{E_pre} are added to the initial prediction values y_{I_pre} to obtain the final runoff prediction result y_{F_pre} , thus achieving BC. The BC formula is as follows, and the modeling process is illustrated in Fig. 3(c).

$$y_{F_pre} = y_{I_pre} + y_{E_pre} \quad (12)$$

This BC mechanism effectively compensates for systematic deviations between the initial predictions and the observed runoff by modeling residual errors, thereby enhancing the overall accuracy and reliability of the final runoff prediction results.

Point prediction model: WVMD-TCN-BiGRU-BC

To improve the accuracy of monthly runoff prediction, this study proposes a multi-strategy hybrid forecasting framework, referred to as the WVMD-TCN-BiGRU-BC model, as illustrated in Fig. 3.

This framework integrates optimized signal decomposition, deep learning based prediction, and bias correction within a unified modeling pipeline.

Specifically, the WVMD method is first employed to decompose the original monthly runoff series into multiple IMFs, thereby reducing the variability and complexity of the original signal. Subsequently, the WOA is applied to optimize the key hyperparameters of the TCN-BiGRU model. The optimized TCN-BiGRU predictor is then used to forecast each decomposed component independently. The predicted components are aggregated to obtain a preliminary runoff prediction, after which a BC strategy is applied to generate the final monthly runoff prediction.

The overall modeling procedure is summarized as follows:

- Decompose the original runoff series into IMFs using the WVMD method.
- Determine appropriate lag times based on the autocorrelation function (ACF) and partial autocorrelation function (PACF), and incorporate them into the prediction model.
- Split the dataset into training and testing subsets and perform data normalization.
- Use the WOA optimized TCN-BiGRU model to predict each decomposed component, followed by inverse normalization of the predicted results.
- Aggregate the predictions of all components to obtain the initial runoff prediction.
- Apply the BC strategy to produce the final runoff prediction results.

Probabilistic runoff interval prediction based on kernel density Estimation (KDE) Determination of KDE bandwidth

Within the probabilistic forecasting framework, Kernel Density Estimation (KDE) is employed to approximate the probability distribution of runoff prediction errors, which serves as the basis for constructing prediction intervals. KDE is a non-parametric technique for estimating the probability density function (PDF) of a random variable from a finite sample set. Given a set of runoff prediction errors $X = \{x_1, x_2, \dots, x_m\}$ independently and identically distributed (i.i.d.) from an unknown distribution $f(x)$, the kernel density estimator $\hat{f}(x)$ is expressed as:

$$\hat{f}(x) = \frac{1}{mh} \sum_{i=1}^m K\left(\frac{x - x_i}{h}\right) \quad (13)$$

where h represents the bandwidth and $K(\cdot)$ is kernel function. Once the sample set is fixed, the performance of KDE is jointly influenced by the choice of kernel function and bandwidth. In practice, however, bandwidth selection plays a dominant role in determining estimation accuracy.

Fig. 4(a) and (b) evaluate the fitting performance of KDE against the standard normal distribution.

$N(0,1)$ by separately investigating the effects of kernel functions and bandwidth selection. The analysis indicates that, compared to the bandwidth, the kernel function has a smaller impact on the kernel density estimation, and the bandwidth is the key factor determining the fitting performance.

To quantitatively assess the accuracy of kernel density estimation, the Integrated Squared Error (ISE) is defined as:

$$ISE(x) = \int \left(\hat{f}(x) - f(x) \right)^2 dx \quad (14)$$

since the true density $f(x)$ is generally unknown, its expectation Mean Integrated Squared Error (MISE) is often minimized instead. Through asymptotic analysis, MISE can be approximated by the Asymptotic Mean Integrated Squared Error (AMISE). Minimizing the AMISE leads to the optimal bandwidth selection:

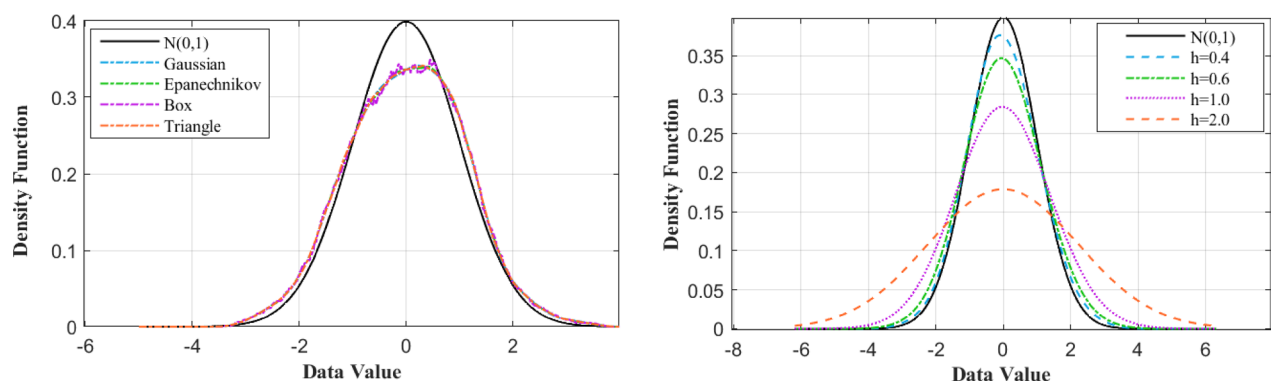


Fig. 4. KDE of the $N(0,1)$ under different kernel functions and bandwidth settings.

$$h_{AMISE} = \left(\frac{\int K^2(x) dx}{h \left(\int x^2 K(x) dx \int (\hat{f}(x))^2 dx \right)} \right)^{1/5} \quad (15)$$

In practical implementations, the Rule of Thumb (ROT) method provides a simple and effective way to approximate h_{AMISE} for Gaussian kernels without requiring explicit knowledge of the true density $\hat{f}(x)$:

$$h_{ROT} = 1.06 \cdot \sigma \cdot n^{-\frac{1}{5}} \quad (16)$$

where σ denotes the standard deviation of the sample data.

Determining the upper and lower bound of the KDE prediction interval

In the probabilistic forecasting stage, the proposed WVMD-TCN-BiGRU-BC model generates point predictions of the runoff series, while KDE is employed to model the probabilistic distribution of the prediction residuals. As the prediction intervals are constructed entirely based on the residuals of the point prediction model, the reliability and sharpness of the interval forecasts are inherently and directly determined by the accuracy of the point predictions, establishing a tight coupling between point and interval prediction stages.

The prediction error between the predicted values $\hat{y}_t$ and the true values y_t is defined as:

$$E = \hat{y}_t - y_t \quad (17)$$

Based on the estimated PDF of the error series, the cumulative distribution function (CDF) can be obtained by integration. Compared with the PDF, the CDF provides the probability that the prediction error is less than or equal to a given value and offers a clearer characterization of the tail behavior of the distribution, which is particularly important for uncertainty quantification and extreme runoff assessment. Therefore, in this study, the CDF is adopted to construct prediction intervals at different confidence levels. The CDF is calculated as:

$$F(x) = \int_{-\infty}^x f(t) dt \quad (18)$$

where $f(x)$ denotes the KDE-based probability density function of the prediction errors.

For a given confidence level $1 - \alpha$, the upper and lower bounds of the prediction error interval can be determined by the inverse function $F^{-1}(\cdot)$ of the CDF:

$$\begin{cases} Lb_{ROT} = F^{-1}\left(\frac{\alpha}{2}\right) \\ Ub_{ROT} = F^{-1}\left(1 - \frac{\alpha}{2}\right) \end{cases} \quad (19)$$

By combining the point prediction values with the estimated prediction error interval, the lower and upper bounds of the runoff prediction interval at the given confidence level can be obtained as

$$\begin{cases} Lb = \hat{y}_t + Lb_{ROT} \\ Ub = \hat{y}_t + Ub_{ROT} \end{cases} \quad (20)$$

Prediction error evaluation metrics

Deterministic point prediction evaluation metrics

To quantitatively evaluate the deterministic point prediction performance of different models, four evaluation metrics are adopted: Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE), the Coefficient of Determination (R^2), and the Ratio of Performance to Deviation (RPD). These metrics jointly assess prediction accuracy, goodness of fit, and residual dispersion from complementary perspectives.

For RMSE and MAPE, smaller values indicate higher prediction accuracy, whereas larger values of R^2 and RPD correspond to better model performance. The evaluation metrics are defined as follows:

$$\left\{ \begin{array}{l} RMSE = \sqrt{\frac{1}{m} \sum_{t=1}^m (y_t - \hat{y}_t)^2} \\ MAPE = \frac{100\%}{m} \sum_{t=1}^m \left| \frac{y_t - \hat{y}_t}{y_t} \right| \\ R^2 = 1 - \frac{\sum_{t=1}^m (y_t - \hat{y}_t)^2}{\sum_{t=1}^m (y_t - \bar{y}_t)^2} \\ RPD = \frac{\sqrt{\frac{1}{m-1} \sum_{t=1}^m (y_t - \bar{y}_t)^2}}{\sqrt{\frac{1}{m-1} \sum_{t=1}^m (e_t - \bar{e})^2}} \end{array} \right. \quad (21)$$

where y_t and $\hat{y}_t$ denote the true and predicted runoff values at time step t , respectively, $\bar{y}_t$ represents the mean of the observed runoff series, $e_t = \hat{y}_t - y_t$ is predicted residual, $\bar{e}$ is the mean of the residual sequence, and m is the total number of samples.

Uncertainty interval prediction evaluation metrics

Accurately fitting the prediction error distribution is a prerequisite for improving the Interval Prediction accuracy. This paper employs error-based root mean squared error (E_{RMSE}), error-based mean absolute error (E_{MAE}), and the error-based coefficient of determination (E_R^2) to evaluate the accuracy of fitting the probability distribution of point prediction errors using KDE, as defined in Eq. (22).

E_{RMSE} and E_{MAE} quantify the average magnitude of fitting errors from different perspectives, where smaller values indicate a closer match between the estimated and empirical distributions. $t_R^2 E_R^2$ measures the proportion of variance in the empirical distribution that can be explained by the estimated distribution. A larger E_R^2 value, closer to 1, indicates better fitting performance.

$$\begin{cases} E_{MAE} = \frac{1}{n} \times \sum_{i=1}^n |c_i - \hat{c}_i| \\ E_{RMSE} = \sqrt{\frac{1}{n} \times \sum_{i=1}^n (c_i - \hat{c}_i)^2} \\ E_R^2 = 1 - \frac{SS_{res}}{SS_{tot}} \end{cases} \quad (22)$$

where c_i represents the empirical cumulative distribution function value, and $\hat{c}_i$ is the estimated CDF value. The E_R^2 is calculated using the sum of squared residuals ($SS_{res} = \sum_{i=1}^n (c_i - \hat{c}_i)^2$) and the total sum of squares ($SS_{tot} = \sum_{i=1}^n (c_i - \bar{c})^2$), where $\bar{c}$ is the mean of c_i .

For evaluating the performance of interval prediction methods, two key metrics are adopted: Prediction Interval Coverage Probability (PICP) and Prediction Interval Normalized Average Width (PINAW). PICP quantifies the proportion of actual observations that fall within the constructed prediction intervals, reflecting the reliability of interval forecasts.

For each time point, if the true value y_t lies between the predicted lower bound $Lb(t)$ and upper bound $Ub(t)$, the indicator function takes a value of 1; otherwise, it is 0:

$$PICP = \frac{1}{m} \sum_{t=1}^m I\{Lb(t) \leq y_t \leq Ub(t)\} \quad (23)$$

where m represents the sample size, and $I(\cdot)$ is an indicator function, which takes the value of 1 if the condition is true and 0 otherwise.

In contrast, PINAW evaluates the sharpness of the prediction intervals by measuring the average interval width normalized by the range of observed values:

$$PINAW = \frac{1}{m \times s} \sum_{t=1}^m (Ub(t) - Lb(t)) \quad (24)$$

where $s = y_{max} - y_{min}$ is the maximum and minimum values of the true observations, respectively.

An effective interval prediction model is expected to achieve both high coverage probability PICP and narrow interval width PINAW. In practice, however, enhancing interval sharpness by reducing PINAW typically compromises coverage probability. To address this inherent trade-off, this study employs the harmonic mean of PICP and PINAW to define a comprehensive F-score metric, with higher values indicating better overall interval prediction performance.

$$F = \begin{cases} \frac{2 \times PICP \times (\frac{1}{PINAW})}{PICP + (\frac{1}{PINAW})}, & \text{if } PINAW > 0 \\ 0, & \text{otherwise} \end{cases} \quad (25)$$

Study area and data

The study area is situated in the Yangtze River Basin, as shown in Fig. 5. It originates from the Qinghai-Tibet Plateau in China and spans approximately 24.5–35.75.5.75°N and 90.55–122.42.55.42°E in longitude. Historically, this basin has been highly susceptible to hydrological extremes, including frequent floods and droughts, owing to its complex climatic conditions and vast drainage area.

A monthly runoff dataset spanning 540 months (January 1979–December 2023) from both stations was used in this study. To reduce potential non-stationarity, linear detrending was applied to the runoff series using a least-squares fitting method over the training period, as illustrated in Fig. 6(a) and (b). The results reveal a gradual increasing trend throughout the study period, suggesting the existence of long-term variability that may be attributed to climatic change and increasing anthropogenic influences such as reservoir regulation and land-use modification. This non-stationary behavior highlights the necessity of employing advanced decomposition and data-driven models to effectively capture both long-term trends and short-term fluctuations in runoff dynamics.

A preliminary time series analysis was then conducted to further examine stationarity and uncover intrinsic temporal characteristics. The ACF and PACF plots, shown in Fig. 6(c) and (d), indicate a pronounced seasonal pattern with an approximate periodicity of 12 months, which is consistent with the annual hydrological cycle of the Yangtze River Basin driven by monsoon precipitation. Moreover, the truncation of the PACF at the 13th lag

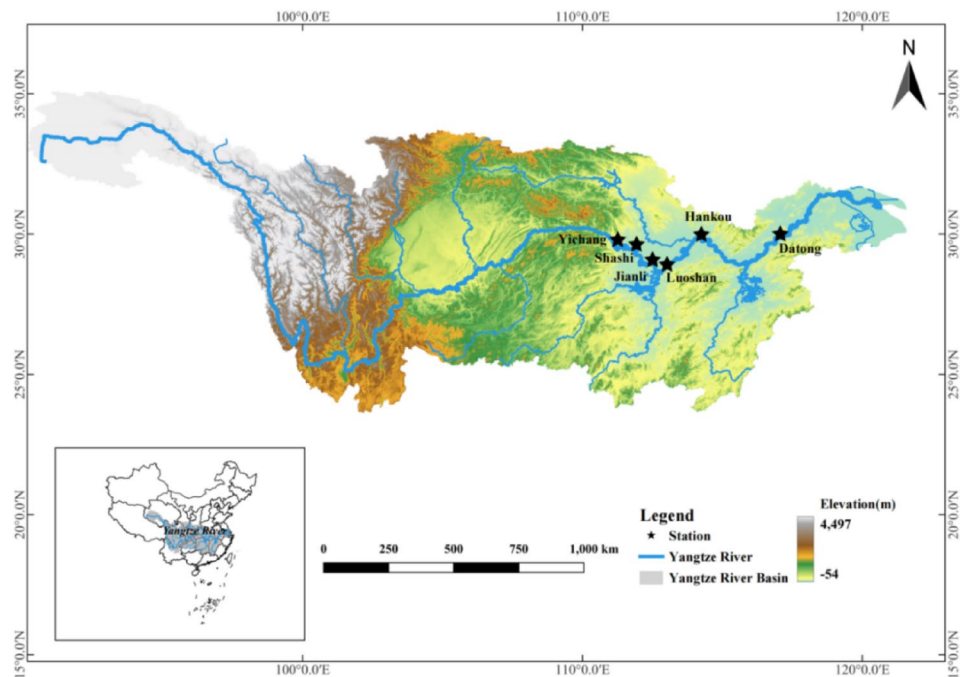


Fig. 5. Location of the Yangtze River Basin and hydrological stations. The map was generated using ArcGIS Pro (version 3.1, <https://www.esri.com/en-us/arcgis/products/arcgis-pro/overview>). Hydrological station locations were manually marked by the authors.

suggests an autoregressive structure of order 12 (AR(12)), implying that runoff values are strongly influenced by hydrological information from the preceding year.

Based on these statistical characteristics, the runoff time series were divided chronologically, with the first 80% of the data used for model training and the remaining 20% reserved for testing to avoid information leakage. Furthermore, a 12-month look-back window was adopted to predict the runoff of the subsequent month, allowing the proposed model to effectively learn seasonal dependencies and inter-annual memory effects embedded in the runoff process.

Results

WVMD process for original runoff

Following the WVMD framework the WOA was employed to optimize the key parameters of VMD, with PE serving as the fitness function. PE is highly sensitive to nonlinear and irregular components in time series data and effectively characterizes the information complexity of decomposed modes. By minimizing PE, WOA identifies the optimal parameter combination that produces simpler, more distinguishable, and less redundant modal components. As a result, the WVMD decomposition yields clearer and more interpretable IMFs, thereby facilitating subsequent prediction modeling.

The parameter search ranges of WVMD were defined for the penalty factor $\alpha \in [100, 2500]$ and the number of modes $K \in [3, 10]$, with a population size of 25 and a maximum of 20 iterations. The convergence behavior of the WOA optimization process is illustrated in Fig. 7(a) and (b), which show the evolution of PE as well as the adaptive trajectories of α and K over successive iterations. The rapid stabilization of PE demonstrates the stable and fast convergence characteristics of WOA. The optimization results indicate that, for Yichang Station, the optimal parameters obtained by WOA are $\alpha = 2117$ and $K = 9$, yielding a minimum PE value of 0.4868. For Jianli Station, the optimal parameter combination is $\alpha = 1958$ and $K = 10$, with the minimum PE value reaching 0.520272. The decomposed IMFs are shown in Fig. 7(c) and (d). This structure allows the subsequent prediction model to effectively capture the intrinsic dynamics of each IMF while reducing the overall model complexity.

Sensitivity analysis of VMD parameters

To investigate the influence of VMD parameter settings on model performance, a sensitivity analysis was conducted by systematically varying the number of modes K and the penalty factor α . In all experiments, the VMD-TCN-BiGRU-BC framework, data partitioning, and prediction model configurations were kept unchanged, and only the VMD parameter pairs (K, α) were manually adjusted. This design ensures that any performance variations can be directly attributed to the VMD parameter settings rather than differences in model structure or training strategy. The test-set results for the Yichang and Jianli stations are summarized in Table 1.

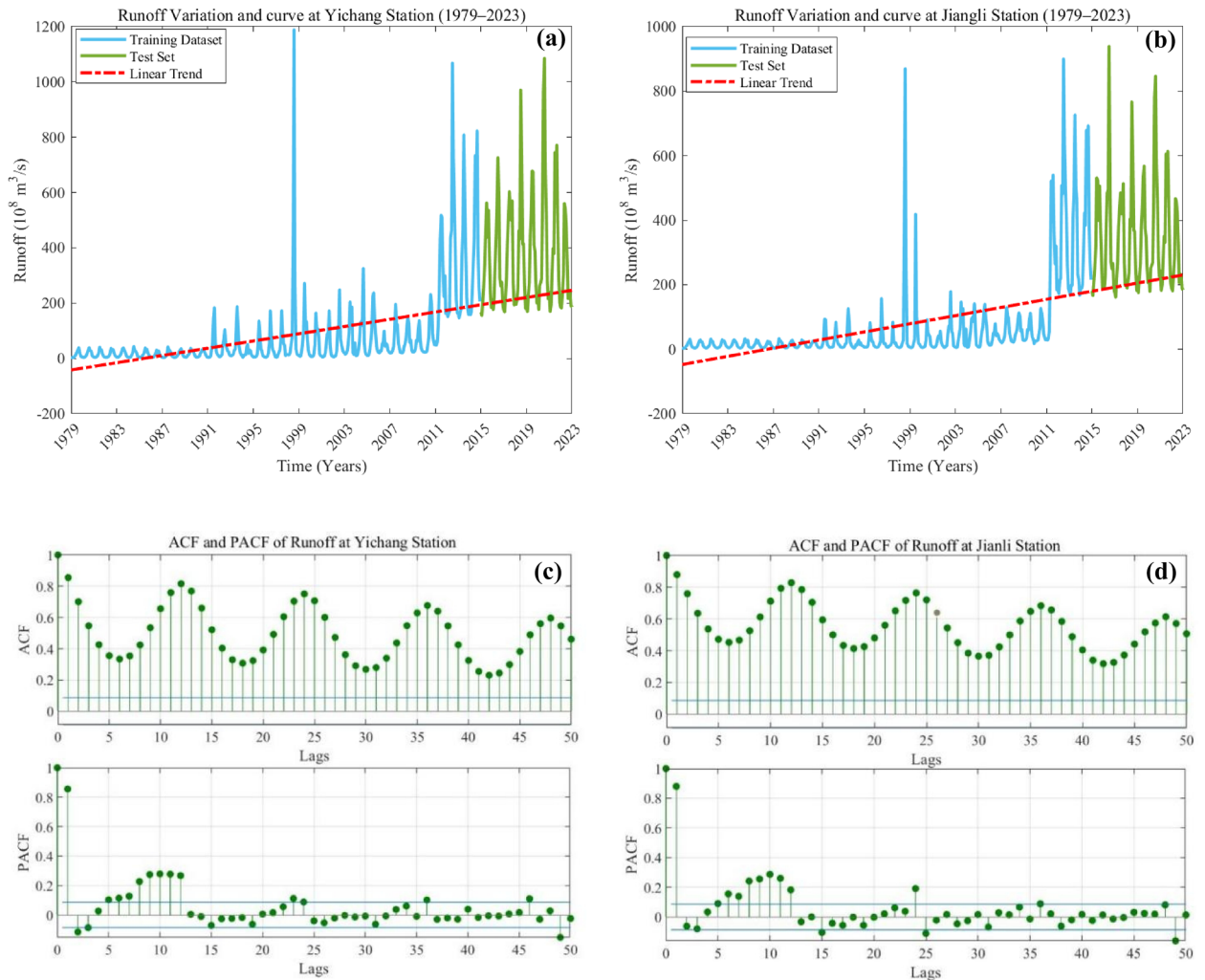


Fig. 6. Original runoff curves and data set statistic analysis.

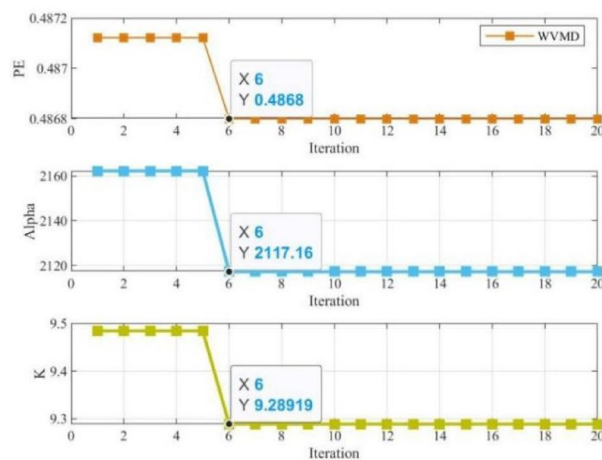
As shown in Table 1, the prediction performance is sensitive to the VMD parameter configuration at both stations. Model accuracy and stability were evaluated using RMSE, MAPE, R^2 , and RPD, where RPD reflects the relative dispersion of errors compared to the variability in observations. In general, variations in α and K lead to noticeable changes in all evaluation metrics, with lower RMSE and MAPE, higher R^2 , and larger RPD values observed under certain parameter combinations, indicating improved accuracy and more stable residual behavior. For Yichang station, RMSE decreases from 55.8220 at (4,500) to 31.6120 at (8,2500), accompanied by an increase in R^2 from 0.9066 to 0.9693. A similar trend is observed at Jianli, where RMSE is reduced to 26.2840 and R^2 reaches 0.9663 across the tested parameter settings.

The sensitivity patterns are further illustrated by the metric curves shown in Fig. 8 for Yichang and Jianli. These curves consistently demonstrate that suitable (K, α) combinations enhance both predictive accuracy and stability, whereas overly small or improperly selected parameters result in degraded performance.

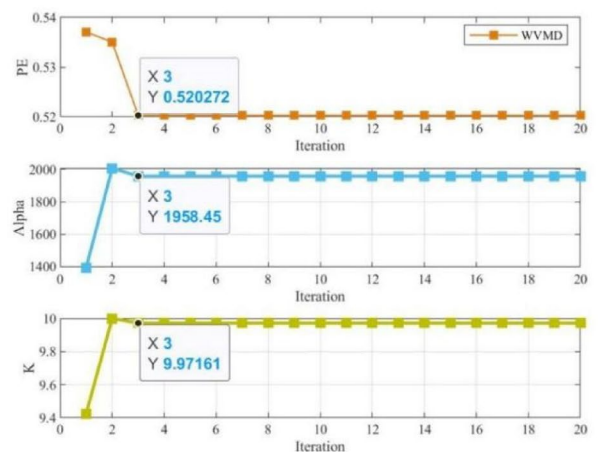
Overall, the sensitivity analysis confirms that the proposed model exhibits a noticeable dependence on the VMD parameter settings, and variations lead to measurable differences in prediction accuracy and residual stability across both stations. These results highlight the importance of examining VMD parameter sensitivity when applying the proposed framework in practical runoff forecasting tasks.

Deterministic point prediction of runoff

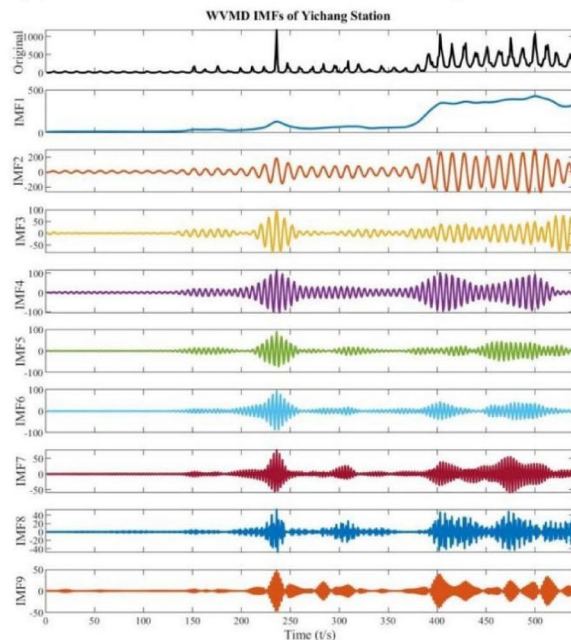
To validate the effectiveness of the proposed WVMD-TCN-BiGRU-BC model, a one-month-ahead forecasting scheme was adopted, in which runoff at time $t + 1$ is predicted using observations from the preceding 12 months. The proposed model was evaluated against a comprehensive set of benchmark methods, including two classical time-series baselines (Persistence and Seasonal-naïve) and several WVMD based deep learning models (CNN, LSTM, TCN, BiGRU, CNN-LSTM, and TCN-BiGRU) as well as Informer and Transformer³⁹. This comparative design inherently forms an implicit ablation analysis, and the corresponding results are summarized in Table 2. For the proposed model, the WOA was employed to optimize key hyperparameters of the TCN-BiGRU network⁴⁰. The WOA was employed to efficiently converge toward the optimal solution, yielding the final configuration [45, 4, 0.1209, 0.0099].



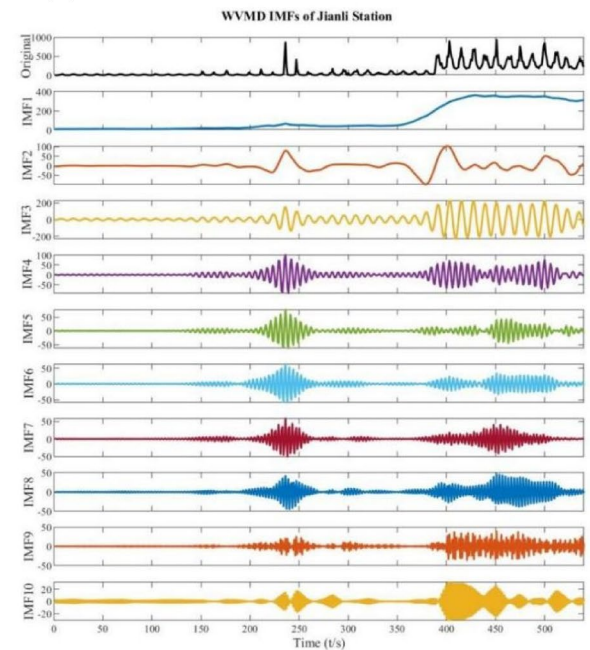
(a) The WVMD results of Yichang Station.



(b) The WVMD results of Jianli Station.



(c) The IMFs of Yichang station.



(d) The IMFs of Jianli station.

Fig. 7. Results of WVMD decomposition of runoff.

(K, α)	Yichang				Jianli			
	RMSE	MAPE	R ²	RPD	RMSE	MAPE	R ²	RPD
(4,500)	55.8220	0.1388	0.9066	3.8801	46.3250	0.1098	0.8980	3.1361
(4, 1000)	50.7110	0.1231	0.9197	4.2152	32.1290	0.0856	0.9485	4.4863
(6,1500)	48.0000	0.1157	0.9291	4.7799	36.6100	0.0836	0.9342	3.9917
(6,2000)	32.1060	0.0768	0.9676	7.3673	28.7350	0.0660	0.9586	4.9497
(8, 2500)	31.6120	0.0789	0.9693	6.5391	26.2840	0.0684	0.9663	5.6459

Table 1. Sensitivity analysis of VMD parameter settings for the VMD-TCN-BiGRU-BC model.

As shown in Table 2, the proposed model consistently outperforms all benchmark methods at both Yichang and Jianli stations across all evaluation metrics. At Yichang, the proposed model achieves an RMSE of 9.5233, with reductions of 68.0% and 74.5% compared to WVMD-TCN-BiGRU and WVMD-CNN-LSTM respectively. Meanwhile, R^2 increases to 0.9973 and RPD rises to 19.8587, about 3.14 times that of WVMD-TCN-BiGRU, indicating a significant reduction in residual dispersion relative to runoff's intrinsic variability. Similar improvements are seen at Jianli station. The proposed model reduces RMSE to 6.6815, a 73.2% decrease from

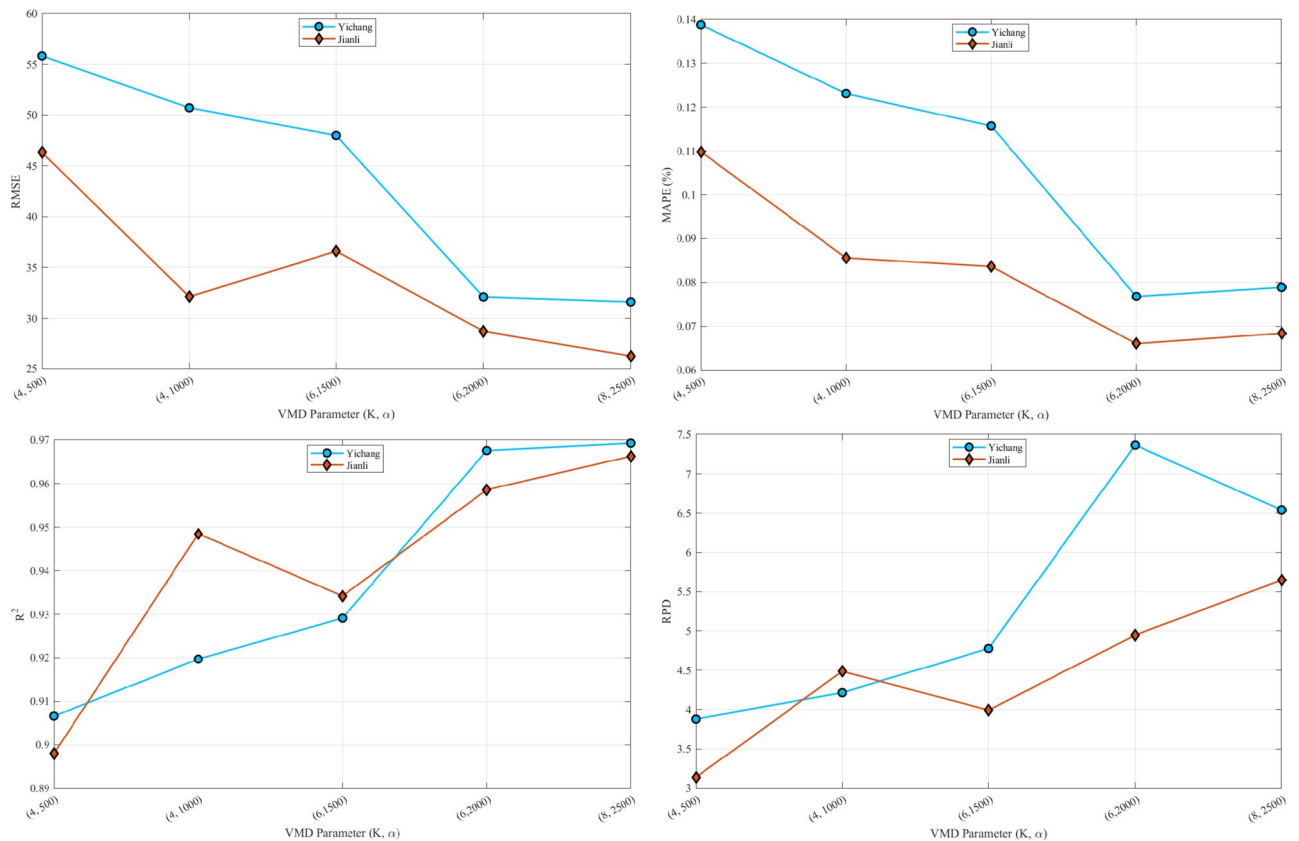


Fig. 8. Sensitivity of prediction performance to VMD parameter settings at the Yichang and Jianli stations.

WVMD-TCN-BiGRU and 74.5% from WVMD-CNN-LSTM. Meanwhile, R^2 rises to 0.9979 and RPD reaches 22.5160, about 3.73 times that of WVMD-TCN-BiGRU, showing highly stable residual behavior. Compared with WVMD-Informer and WVMD-Transformer, the proposed model has clear advantages at both stations, proving the effectiveness of integrating optimized temporal convolution, bidirectional recurrent learning, and bias correction for monthly runoff forecasting.

The point prediction curves shown in Fig. 9(a) and (d) further illustrate the effectiveness of the proposed approach. Compared with other models, the WVMD-based preprocessing enhances prediction smoothness and consistency by mitigating the non-linearity and non-stationarity of the original runoff series. In particular, the proposed model more accurately captures the peak dynamics during the flood season (May–October 2020, samples 63–68), as shown in Fig. 9(c) and (d), which corresponds to periods of extreme runoff in the Yangtze River Basin. This highlights the practical reliability of the proposed framework for high-risk hydrological forecasting scenarios.

Overall, these results demonstrate that the optimized WVMD decomposition provides a robust and informative representation for runoff forecasting, while the subsequent bias-correction module further refines the deterministic predictions by effectively reducing systematic residual errors. The combined decomposition-prediction-correction framework therefore offers a reliable and practically effective solution for high-risk monthly runoff forecasting scenarios.

Runoff error probability distribution fitting and interval prediction

Error distribution characterization and density fitting

To characterize the statistical structure of runoff prediction errors, the residual series obtained from the WVMD-TCN-BiGRU-BC model were first examined through PDF analysis. Three distribution modeling approaches were considered, including the Normal distribution, the Gumbel distribution, and the ROT-KDE. Fig. 10(a) and (b) illustrate the PDF representations of the residuals at Yichang and Jianli stations, respectively. The residual distributions at both stations exhibit slight right skewness and heavy-tailed characteristics (Yichang: skewness = 0.3559, kurtosis = 3.7086; Jianli: skewness = 0.3973, kurtosis = 3.1361), indicating clear deviations from the Gaussian assumption. These structural features revealed by the PDFs suggest that conventional parametric distributions may be inadequate for accurately characterizing the error behavior. Accordingly, the nonparametric ROT-KDE method was adopted to flexibly capture the empirical density structure of the residuals, with bandwidths determined as 10.072 for Yichang and 7.510 for Jianli based on the Rule-of-Thumb criterion.

Building upon the PDF-based error structure analysis, the corresponding CDFs were further investigated to assess the global distributional consistency of different modeling approaches and to support subsequent

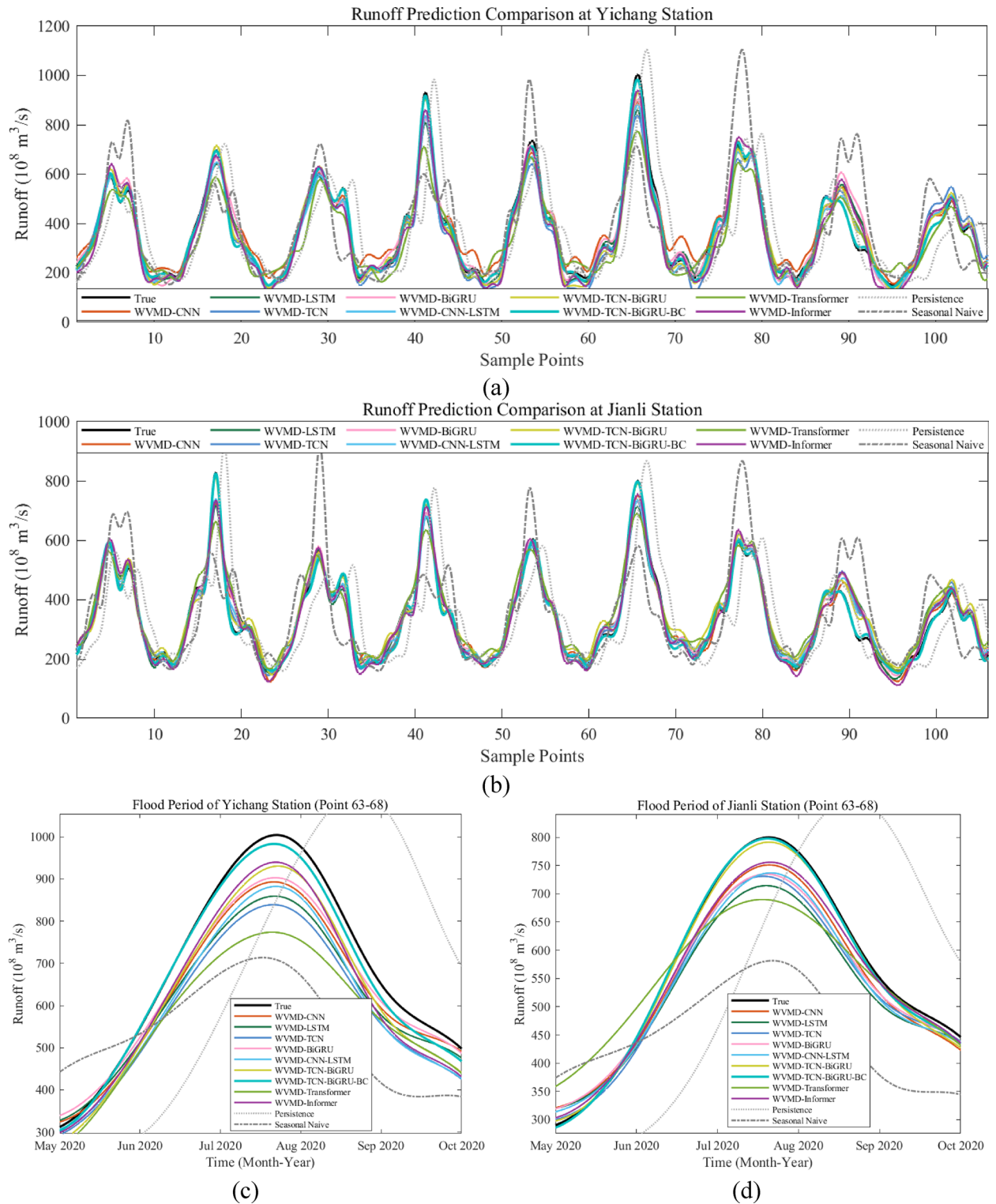


Fig. 9. Point prediction analysis of different prediction models.

uncertainty quantification. Fig. 10(c) and (d) present the empirical CDFs of the residuals together with the fitted CDFs derived from the Normal distribution, Gumbel distribution, and ROT-KDE for the two stations. It can be observed that the ROT-KDE-based CDF exhibits the closest agreement with the empirical CDF over the entire probability range, particularly in the tail regions that are critical for extreme error representation. In contrast, the parametric distributions show noticeable deviations in both central and tail regions, reflecting their limited ability to accommodate skewness and heavy-tailed behavior. This combined PDF-CDF analysis demonstrates that ROT-KDE not only captures the local density characteristics of runoff prediction errors but also provides

Model	Yichang				Jianli			
	RMSE	MAPE	R^2	RPD	RMSE	MAPE	R^2	RPD
Persistence	142.0398	27.1710	0.4621	1.4071	123.9286	24.3992	0.3720	1.3021
Seasonal-naive	147.4259	24.3830	0.4205	1.3555	131.6729	24.0313	0.2911	1.2257
WVMD-Transformer	61.3560	0.1095	0.8872	3.0726	44.2484	0.1001	0.9083	3.4080
WVMD-Informer	42.6120	0.1040	0.9455	4.4210	32.5799	0.0833	0.9503	4.6274
WVMD-CNN	42.9356	0.1142	0.9447	4.3877	29.5778	0.0691	0.9590	5.0960
WVMD-LSTM	39.8584	0.0768	0.9233	3.7254	33.431	0.0720	0.9476	4.5081
WVMD-TCN	45.4645	0.1057	0.9379	4.1410	27.4302	0.0601	0.9647	5.4918
WVMD-BiGRU	43.8936	0.0863	0.9422	4.2926	30.3836	0.0624	0.9567	4.9587
WVMD-CNN-LSTM	37.3336	0.0841	0.9582	5.0461	26.1407	0.0576	0.9680	5.7681
WVMD-TCN-BiGRU	29.7950	0.0744	0.9734	6.3270	24.9495	0.0694	0.9708	6.0380
WVMD-TCN-BiGRU-BC	9.5233	0.0219	0.9973	19.8587	6.6815	0.0202	0.9979	22.5160

Table 2. Error evaluation metric of the point prediction model on the test set.

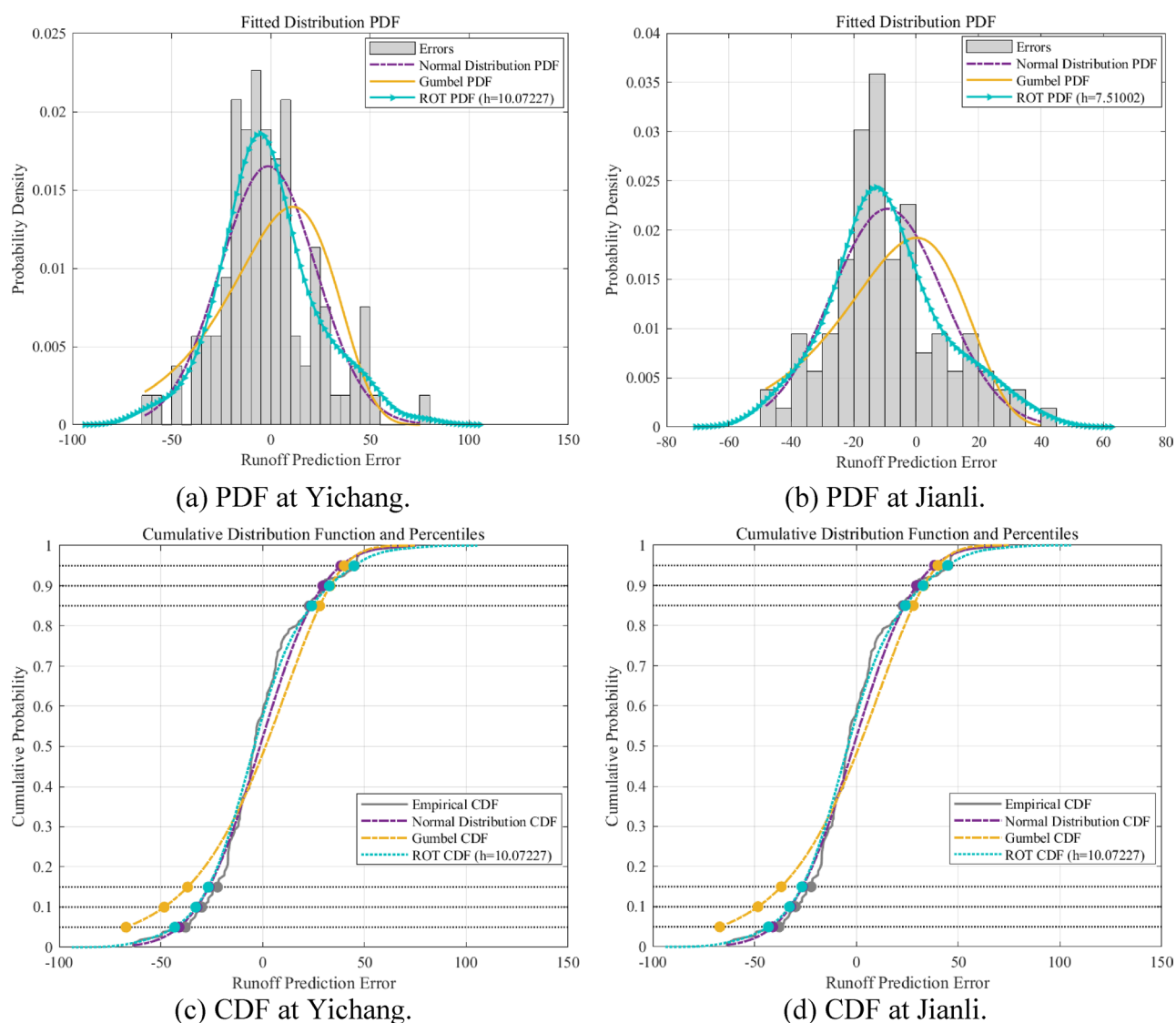


Fig. 10. PDF and CDF analyses of runoff point prediction errors at Yichang and Jianli stations.

a more accurate global cumulative representation, thereby establishing a reliable probabilistic foundation for interval prediction.

The comparative distribution fitting performance was evaluated using three CDF-based metrics, namely E_{MAE} , E_{RMSE} and the coefficient of determination E_{R^2} , as reported in Table 3. These metrics quantify the discrepancies between the empirical CDF and the fitted CDFs obtained from different distribution models. As shown in Table 3, ROT-KDE consistently yields the lowest E_{MAE} and E_{RMSE} and the highest E_{R^2} at both stations. At Yichang station, ROT-KDE reduces E_{MAE} and E_{RMSE} by 28.55% and 32.08% relative to the Normal distribution, and by 65.43% and 64.02% compared with the Gumbel distribution, respectively. Similar improvements are observed at Jianli station, confirming that ROT-KDE provides a more accurate and stable approximation of the residual CDF. Based on the above CDF fitting results, ROT-KDE is adopted for subsequent probabilistic interval construction, enabling uncertainty quantification of runoff predictions at different confidence levels.

Probabilistic interval prediction based on ROT-KDE

Based on the CDF-based error distribution analysis , probabilistic prediction intervals were constructed using the Normal distribution, Gumbel distribution, and ROT-KDE. Prediction intervals were generated at three confidence levels (70%, 80%, and 90%), and the corresponding results for Yichang and Jianli stations are illustrated in Fig. 11, with quantitative evaluations summarized in Table 4.

Interval prediction performance was assessed using PICP, PINAW, F-score, Winkler score, and coverage error. As shown in Table 4, the ROT-KDE-based intervals consistently achieve higher PICP values across all confidence levels at both stations, indicating more reliable coverage of observed runoff. In contrast, the Normal distribution generally produces narrower intervals but fails to meet nominal coverage requirements, particularly at higher confidence levels, while the Gumbel distribution exhibits inferior performance due to its limited ability to capture the asymmetric and heavy-tailed error structure.

Moreover, the interval metrics exhibit a clear confidence-level-dependent pattern. As the confidence level increases, PICP rises accordingly, accompanied by moderate increases in PINAW, demonstrating that the constructed intervals respond appropriately to different confidence requirements. Across all confidence levels, ROT-KDE maintains the most favorable balance between coverage and interval width, yielding the highest F-scores among the compared methods.

Overall, these results confirm that the ROT-KDE-based interval prediction approach provides stable and statistically consistent uncertainty quantification for runoff forecasting, effectively complementing deterministic point prediction results.

Discussion and conclusion

Discussion

Based on the comprehensive point prediction evaluation in Table 2, this study further examines whether the superiority of the proposed framework can be consistently preserved when extended from deterministic forecasting to probabilistic interval prediction. Accordingly, the first four simple baseline models were excluded, and the remaining seven DL-based models were uniformly combined with the proposed ROT-KDE strategy to construct prediction intervals at the 90% confidence level. The corresponding quantitative results are summarized in Table 5, with representative interval prediction curves illustrated in Fig. 12.

The results clearly show that point prediction accuracy plays a decisive role in interval prediction performance. Models with higher point prediction accuracy generally produce smaller and more stable residuals, which directly facilitates more accurate density estimation and leads to narrower, better-calibrated prediction intervals. In contrast, inferior point prediction performance results in dispersed residual distributions, inevitably degrading interval sharpness even when the same probabilistic method is applied.

Benefiting from its superior deterministic forecasting capability, the proposed WVMD-TCN-BiGRU -BC-ROT-KDE framework achieves the best overall interval prediction performance at both stations. As reported in Table 5, compared with other DL based models, it reduces E_{MAE} and E_{RMSE} by approximately 15%-40%, while further decreasing PINAW by about 20%-35%, with PICP values remaining at or above the nominal confidence level. Consequently, the comprehensive F score is improved by roughly 5%-10%, indicating a more favorable balance between interval reliability and sharpness. These quantitative advantages are also consistently reflected in Fig. 12(a)-(b), where the prediction intervals generated by the proposed model are visibly more compact and stable, yet consistently encompass the observed runoff series.

Overall, these results demonstrate that the proposed KDE based probabilistic framework effectively translates improvements achieved at the point prediction stage into more informative and reliable prediction intervals. By accurately modeling the error distribution derived from high-quality point forecasts, the proposed approach enhances uncertainty quantification and provides a robust tool for probabilistic runoff prediction.

Distribution Function	Yichang			Jianli		
	E_{MAE}	E_{RMSE}	E_{R^2}	E_{MAE}	E_{RMSE}	E_{R^2}
Normal	0.0339	0.0434	0.9677	0.0326	0.0413	0.9791
Gumbel	0.0700	0.0821	0.9206	0.0662	0.0780	0.9284
ROT-KDE	0.0242	0.0295	0.9897	0.0236	0.0282	0.9906

Table 3. CDF-based evaluation of runoff point prediction error distribution fitting.

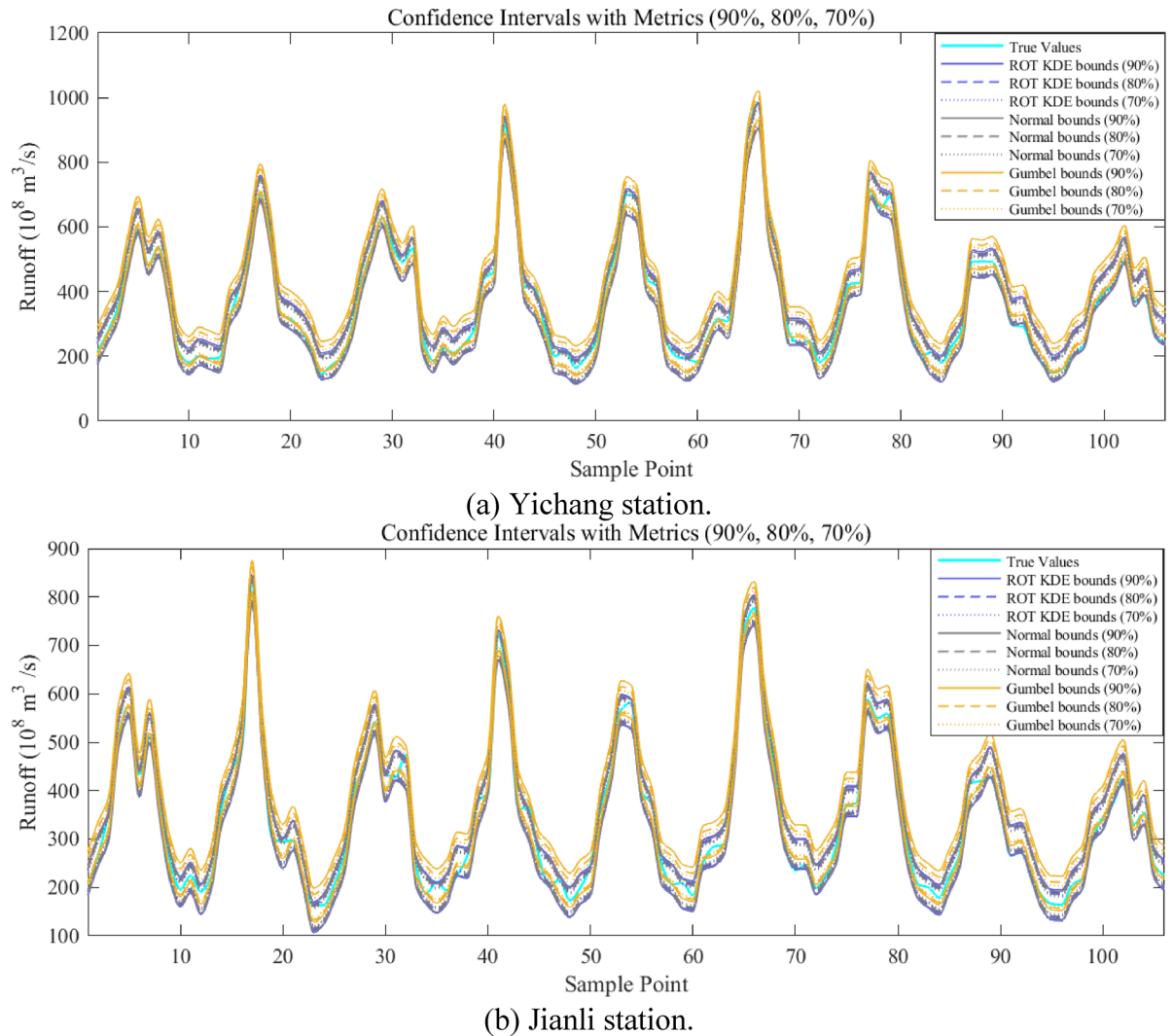


Fig. 11. Probabilistic runoff prediction intervals at different confidence levels.

Conclusion

This study proposed a probabilistic runoff forecasting framework that integrates optimized signal decomposition, DL models, BC mechanism, and uncertainty quantification. Based on comprehensive experiments conducted at Yichang and Jianli stations, the main conclusions corresponding to the stated contributions are summarized as follows:

- i. Effectiveness of WVMD-based decomposition.
- ii. The WVMD method effectively decomposes non-stationary runoff series into high quality IMFs, improving data stationarity and noise separation. This decomposition provides more informative inputs for subsequent modeling and enhances the overall predictability of runoff time series.
- iii. Performance of the optimized hybrid deep learning model.
- iv. The proposed WVMD-TCN-BiGRU architecture, optimized via the WOA, successfully combines the strengths of temporal convolution and bidirectional recurrent learning. This hybrid structure captures long term temporal dependencies and bidirectional dynamics, leading to improved deterministic runoff forecasting performance.
- v. Contribution of the BC mechanism.
- vi. The incorporation of the BC strategy significantly mitigates systematic prediction errors. Compared with the corresponding WVMD-TCN-BiGRU model, the proposed WVMD-TCN-BiGRU-BC framework reduces RMSE by 68.0% and 73.2% at the two stations, respectively, demonstrating substantial improvements in accuracy and robustness.
- vii. Reliability of KDE-based probabilistic interval prediction.
- viii. By applying ROT-KDE to the prediction residuals of the proposed model, reliable probabilistic prediction intervals are constructed. Compared with parametric distributions, the KDE-based approach achieves

Station	Distribution Function	Confidence Level	PICP	PINAW	F	Winkler	Coverage Error
Yichang	Normal	90%	0.8774	0.0896	1.6268	92.5227	-0.0226
		80%	0.7547	0.0698	1.4339	82.9173	-0.0453
		70%	0.6604	0.0565	1.2733	72.9703	-0.0396
	Gumbel	90%	0.7453	0.1104	1.3773	169.4625	-0.1547
		80%	0.6321	0.0837	1.2006	127.7803	-0.1679
		70%	0.5189	0.0667	1.0030	106.7261	-0.1811
	ROT-KDE	90%	0.9340	0.0991	1.7097	89.3280	0.0340
		80%	0.8113	0.0783	1.5257	79.6299	0.0113
		70%	0.7264	0.0629	1.3894	70.0355	0.0264
Jianli	Normal	90%	0.8868	0.0854	1.6487	68.7992	-0.0132
		80%	0.7453	0.0666	1.4201	60.6965	-0.0547
		70%	0.6415	0.0538	1.2402	56.2465	-0.0585
	Gumbel	90%	0.7642	0.1036	1.4162	134.7405	-0.1358
		80%	0.6604	0.0785	1.2556	99.4015	-0.1396
		70%	0.5377	0.0626	1.0405	82.2638	-0.1623
	ROT-KDE	90%	0.9434	0.0934	1.7341	67.9227	0.0434
		80%	0.8208	0.0755	1.5457	60.2965	0.0208
		70%	0.7019	0.0614	1.4123	54.5202	0.0019

Table 4. Evaluation results of probabilistic runoff prediction intervals at different confidence levels.

Station	Model	E_{MAE}	E_{RMSE}	$E_R^2 E_{R2}$	PICP	PINAW	F
Yichang	WVMD-CNN-ROT-KDE	0.1918	0.2825	0.9155	0.9340	0.1702	1.6117
	WVMD-LSTM-ROT-KDE	0.2786	0.4666	0.9004	0.9057	0.1622	1.5794
	WVMD-TCN-ROT-KDE	0.1993	0.2762	0.9285	0.9057	0.1790	1.5586
	WVMD-BiGRU-ROT-KDE	0.2304	0.3640	0.9119	0.9245	0.1732	1.5938
	WVMD-CNN-LSTM-ROT-KDE	0.2467	0.3437	0.9060	0.9151	0.1366	1.6268
	WVMD-TCN-BiGRU-ROT-KDE	0.1543	0.2016	0.9208	0.9340	0.1227	1.6758
	WVMD-TCN-BiGRU-BC-ROT-KDE	0.1548	0.1983	0.9551	0.9340	0.0991	1.7097
Jianli	WVMD-CNN-ROT-KDE	0.0979	0.1512	0.9816	0.9434	0.1476	1.6562
	WVMD-LSTM-ROT-KDE	0.2295	0.4090	0.9050	0.9151	0.1668	1.5878
	WVMD-TCN-ROT-KDE	0.1006	0.1415	0.9861	0.9151	0.1337	1.6306
	WVMD-BiGRU-ROT-KDE	0.2552	0.3972	0.9041	0.9340	0.2356	1.5311
	WVMD-CNN-LSTM-ROT-KDE	0.1483	0.2120	0.9715	0.9245	0.1281	1.6533
	WVMD-TCN-BiGRU-ROT-KDE	0.0730	0.0945	0.9855	0.9340	0.0954	1.7151
	WVMD-TCN-BiGRU-BC-ROT-KDE	0.0625	0.0722	0.9869	0.9434	0.0934	1.7341

Table 5. Interval prediction results of DL models using ROT-KDE at 90% confidence level.

higher coverage quality and yields an approximately 5% improvement in the composite Fscore at the 90% confidence level, providing more credible uncertainty quantification for runoff forecasting.

Although the proposed framework achieves significant improvements in both point and interval prediction accuracy, several limitations remain. First, the multi-method ensemble structure increases model complexity and computational cost, which may restrict real-time applications. Second, the model performance depends on the quality and stationarity of input data, and its generalization ability across different basins still requires further validation. Third, the bias correction and KDE based interval prediction processes add additional modeling steps, which may lead to overfitting if not properly regularized. Future research will focus on simplifying the architecture, improving computational efficiency, and testing the model's transferability to other hydrological environments.

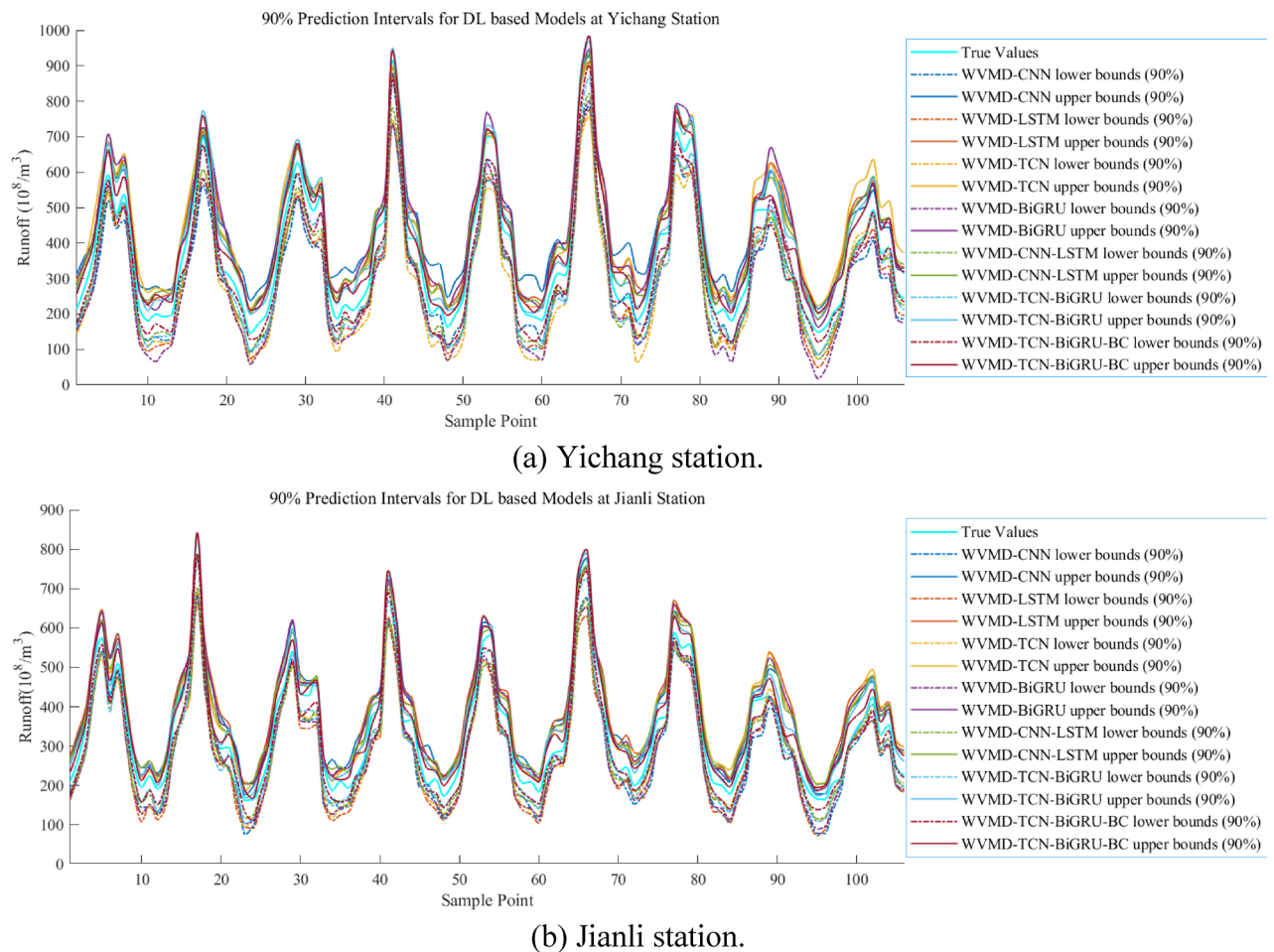


Fig. 12. Comparison of probabilistic interval prediction results of DL models using ROT-KDE at the 90% confidence level.

Data availability

The datasets used and analysed during the current study available from the corresponding author on reasonable request.

Received: 10 August 2025; Accepted: 22 December 2025

Published online: 05 March 2026

References

1. Zhang, S., Ren, G., Zhan, Y., Zhang, C. & Ren, Y. Extreme precipitation changes over the Yangtze river basin in 1901–2020. *Climate Res.* **90**, 59–76 (2023).
2. Zhang, Q., Peng, J., Xu, C. Y. & Singh, V. P. Spatiotemporal variations of precipitation regimes across Yangtze river Basin, China. *Theoret. Appl. Climatol.* **115**, 703–712 (2014).
3. Gong, L., Zhang, X., Liu, J. & Gui, H. Exploring the influence of urban agglomeration on extreme precipitation: evidence from the middle reaches of the Yangtze River, China. *J. Hydrology: Reg. Stud.* **55**, 101932 (2024).
4. Dey, P. & Mishra, A. Separating the impacts of climate change and human activities on streamflow: A review of methodologies and critical assumptions. *J. Hydrol.* **548**, 278–290 (2017).
5. Hosseini, F., Prieto, C. & Álvarez, C. An explainable AI approach for interpreting regionally optimized deep neural networks in hydrological prediction. *J. Hydrol.*, 133689. (2025).
6. Sit, M., Demiray, B. Z., Xiang, Z., Ewing, Y. S. & Demir, I. A comprehensive review of deep learning applications in hydrology and water resources. *Water Sci. Technol.* **82** (12), 2635–2670. <https://doi.org/10.2166/wst.2020.369> (2020).
7. Zhang, X., Peng, Y., Zhang, C. & Wang, B. Are hybrid models integrated with data preprocessing techniques suitable for monthly streamflow forecasting? Some experiment evidences. *J. Hydrol.* **530**, 137–152 (2015).
8. Vishwakarma, D. K. et al. *Improving Streamflow Forecasting Efficiency Using Signal Decomposition Approaches* 1–34 (Water Resources Management, 2025).
9. Sattari, A., Foroumandi, E., Gavahi, K. & Moradkhani, H. A probabilistic machine learning framework for daily extreme events forecasting. *Expert Syst. Appl.* **265**, 126004 (2025).
10. Yuan, R. et al. Daily runoff forecasting using ensemble empirical mode decomposition and long short-term memory. *Front. Earth Sci.* **9**, 621780 (2021).
11. Huang, S. et al. Runoff prediction of irrigated paddy areas in Southern China based on EEMD-LSTM model. *Water* **15** (9), 1704 (2023).

12. Dragomiretskiy, K. & Zosso, D. Variational mode decomposition. *J. IEEE Trans. signal. Process.* **62**, 531–544 (2013).
13. Jing, X., Luo, J., Zhang, S. & Wei, N. Runoff forecasting model based on variational mode decomposition and artificial neural networks. *Math. Biosci. Eng.* **19** (2), 1633–1648 (2022).
14. Wang, X., Wang, Y., Yuan, P., Wang, L. & Cheng, D. An adaptive daily runoff forecast model using VMD-LSTM-PSO hybrid approach. *Hydrol. Sci. J.* **66** (9), 1488–1502 (2021).
15. Li, B. J., Sun, G. L., Liu, Y., Wang, W. C. & Huang, X. D. Monthly runoff forecasting using variational mode decomposition coupled with Gray Wolf optimizer-based long short-term memory neural networks. *Water Resour. Manage.* **36** (6), 2095–2115 (2022).
16. Wang, W. C. et al. Evaluating the performance of several data preprocessing methods based on GRU in forecasting monthly runoff time series. *Water Resour. Manage.*, 1–18. (2024).
17. Bai, S., Kolter, J. Z. & Koltun, V. An empirical evaluation of generic convolutional and recurrent networks for sequence modeling. *ArXiv Preprint arXiv:180301271*. (2018).
18. Wang, H. et al. Dynamic Displacement Response Prediction of Isolation Bearings Based on Multiscale Decomposition and Combined Method. Available at SSRN 4969477.
19. Zhang, L. & Xue, G. Short-Term Heat Load Forecasting Based on Ceemd and a Hybrid Idbo-Tcn-Bigrun Network. Available at SSRN 5030114.
20. Chen, H., Huang, H., Zheng, Y. & Yang, B. A load forecasting approach for integrated energy systems based on aggregation hybrid modal decomposition and combined model. *Appl. Energy*. **375**, 124166 (2024).
21. Li, Z. & Ju, Y. Short-Term Photovoltaic Power Prediction Based on VMD-mRMR and TCN-BIGRU-ATTENTION. In 2024 9th International Symposium on Computer and Information Processing Technology (ISCIPIT) (pp. 386–389). IEEE. (2024), May.
22. Zhang, Y., Zhang, L., Sun, D., Jin, K. & Gu, Y. Short-term wind power forecasting based on VMD and a hybrid SSA-TCN-BiGRU network. *Appl. Sci.* **13** (17), 9888 (2023).
23. Zou, Z., Wang, J., Zhang, E. N., Wang, C., Jiang, E. & Z., & Short-term power load forecasting: an integrated approach utilizing variational mode decomposition and TCN-BiGRU. *Energies* **16** (18), 6625 (2023).
24. Danandeh Mehr, A., Ghadimi, S., Marttila, H. & Torabi Haghighi, A. A new evolutionary time series model for streamflow forecasting in boreal lake-river systems. *Theoret. Appl. Climatol.* **148** (1), 255–268 (2022).
25. Farzin, S. & Valikhan Anaraki, M. Modeling and predicting suspended sediment load under climate change conditions: a new hybridization strategy. *J. Water Clim. Change*. **12** (6), 2422–2443 (2021).
26. Nasiri, J. & Farzin Modarres Khiyabani A Whale optimization algorithm (WOA) approach for clustering. *Cogent Math. Stat.* **5** (1), 1483565 (2018).
27. Zhao, H., Zhou, Z. & Zhang, P. Forecasting of the Short-Term electricity load based on WOA-BiLSTM. *Int. J. Pattern Recognit. Artif. Intell.*, 37(11). (2023).
28. Chen, J. Ultra-short-term Wind Power Prediction Research Based on WOA-BiLSTM-AM Model. In 2023 2nd International Conference on Advanced Electronics, Electrical and Green Energy (AEEGE) (pp. 100–104). IEEE. (2023), May.
29. Ji, C., Peng, T., Zhang, C., Hua, L. & Sun, W. An Integrated Framework of GRU Based on Improved Whale Optimization Algorithm for Flood Prediction. (2021).
30. Ngai, S. T., Juneng, L., Tangang, F., Chung, J. X., Supari, S., Salimun, E., ... Gunawan, D. (2022). Projected mean and extreme precipitation based on bias-corrected simulation outputs of CORDEX Southeast Asia. *Weather and Climate Extremes*, 37, 100484.
31. Dao, V. et al. Bias correction of satellite precipitation Estimation using deep neural networks and topographic information over the Western US. *J. Geophys. Research: Atmos.*, **130** (4), (2025). e2024JD042181.
32. Zhang, Y., Pan, G., Zhao, Y., Li, Q. & Wang, F. Short-term wind speed interval prediction based on artificial intelligence methods and error probability distribution. *Energy. Conv. Manag.* **224**, 113346 (2020).
33. Yang, X., Ma, X., Kang, N. & Maihemuti, M. Probability interval prediction of wind power based on KDE method with rough sets and weighted Markov chain. *IEEE Access.* **6**, 51556–51565 (2018).
34. Klotz, D., Kratzert, F., Gauch, M., Keefe Sampson, A., Brandstetter, J., Klambauer, G., ... Nearing, G. (2022). Uncertainty estimation with deep learning for rainfall-runoff modeling. *Hydrology and Earth System Sciences*, 26(6), 1673–1693.
35. Khaldi, R., El Afia, A. & Chiheb, R. Forecasting of BTC volatility: comparative study between parametric and nonparametric models. *Progress Artif. Intell.* **8** (4), 511–523 (2019).
36. Mirjalili, S. *Andrew Lewis Whale Optim. Algorithm Adv. Eng. Softw.* **95**: 51–67. (2016).
37. Ni, C., Marsani, M. F., Shan, F. P. & Zou, X. Flood prediction with optimized gated recurrent unit-temporal convolutional network and improved KDE error Estimation. *AIMS Math.* **9** (6), 14681–14696 (2024).
38. Abid, F., Alam, M., Alamri, F. S. & Siddique, I. Multi-directional gated recurrent unit and convolutional neural network for load and energy forecasting: A novel hybridization. *AIMS Math.* **8** (9), 19993–20017 (2023).
39. Fan, J., Yu, G. A., Zhao, M. & Zong, H. *Addressing multi-scale Temporal Variability: Deep Integration and Application of the CNN and Transformer Model in Monthly Streamflow Prediction* 128658 (Expert Systems with Applications, 2025).
40. Wang, S., Shao, C., Xu, S., Yang, X. & Yu, H. MSFSS: A whale optimization-based multiple sampling feature selection stacking ensemble algorithm for classifying imbalanced data. *learning*, 26, 31. (2024).

Author contributions

Hong Ma and Muhammad Fadhil Marsani conceived the study and developed the overall methodology. Hong Ma, Muhammad Fadhil Marsani, and Caiyan Long designed the experiments and conducted the data analysis. Mohd. Asyraf Mansor and Mohd Shareduwan Mohd Kasihmuddin contributed to data collection and preprocessing. Hong Ma and Caiyan Long implemented the deep learning models and optimization algorithms. Muhammad Fadhil Marsani and Mohd Shareduwan Mohd Kasihmuddin performed the bias correction strategy analysis. Hong Ma and Muhammad Fadhil Marsani drafted the original manuscript. All authors participated in result interpretation, manuscript revision, and approved the final version of the manuscript.

Funding

The authors received no specific funding for this work.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to M.F.M.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

© The Author(s) 2025